QueueLite

Appointment and Queue Management System

Software Engineering Fundamentals
Course Project Report

Prepared by
Retaj Majed Alotaibi
Shamayil Khouj
Elham Alzahrani
Sahar Hawsawi

Supervised by
Dr. Fawziah Ghazawi

Contents

Team Work Distribution

1. Planning Stage

1.1 Project Idea and Justification · 1.2 Problem Statement · 1.3 Scope of Work · 1.4 Project Timeline

2. Analysis Stage

2.1 Functional Requirements · 2.2 Non-Functional Requirements · 2.3 Target Users and Their Needs · 2.4 User Personas · 2.5 Use Case Diagram · 2.6 Risk Assessment and Mitigation

3. Design Stage

3.1 UML Class Diagram · 3.2 System Architecture · 3.3 UI Wireframes

4. Testing Stage

4.1 Test Scenarios · 4.2 Test Cases · 4.3 Bug Reports

References

Team Work Distribution

Team Member	Responsibilities
Retaj Majed Alotaibi	• Wrote the Project Idea and Justification section • Explained the importance of the system • Defined the target users
Shamayil Khouj	• Wrote the Problem Statement section • Identified the scheduling issues • Explained the impact on clinics
Elham Alzahrani	• Defined the In Scope / Out of Scope items • Organized the features and user roles • Clarified the project boundaries
Sahar Hawsawi	• Prepared the project timeline table • Organized the milestones • Numbered and formatted the references and the document

1. Planning Stage

1.1 Project Idea and Justification

QueueLite is a lightweight appointment and queue management system designed for small clinics, salons, and consultation centers. The system allows clients to book, modify, or cancel appointments, while receptionists manage the daily schedule and managers monitor daily reports.

The idea was chosen because many small clinics still rely on manual booking methods such as notebooks or messaging apps, which often result in lost appointments, double bookings, and poorly organized schedules. QueueLite provides a simple digital solution tailored to small businesses.

1.2 Problem Statement

Small clinics and service centers frequently experience scheduling issues because they rely on manual systems. Common problems include double bookings, missed appointments, last-minute cancellations without notice, and a lack of performance tracking. These issues reduce productivity and negatively affect customer satisfaction.

QueueLite addresses these challenges by centralizing appointment management, preventing conflicting bookings, and sending automated email reminders before each appointment.

1.3 Scope of Work

In Scope	Out of Scope
• User roles: Client, Receptionist, Manager • Client registration, login and password reset • Role-based access control • Book, modify, and cancel appointments • Prevention of double booking for the same time slot • Appointment status management (Scheduled / Completed / Cancelled / No-Show) • Automated email reminders before appointments • Daily reports (bookings, cancellations, attendance rate)	• Online payment integration • SMS or WhatsApp notifications • AI-based scheduling • Multi-branch support • Integration with hospital systems

1.4 Project Timeline (Week 1 – Week 9)

Week	Stage / Milestone
Week 1	Requirements and Analysis
Week 2	System Design (UML Class Diagram, Use Case Diagram, Architecture, UI Wireframes)
Week 3	Database Implementation
Week 4	Authentication Module (Registration, Login and Roles)
Week 5	Appointment Module (Book / Modify / Cancel)
Week 6	Email Reminder Integration
Week 7	Reporting Module
Week 8	Testing and Debugging
Week 9	Final Documentation and Presentation

2. Analysis Stage

2.1 Functional Requirements

Requirements are written as "shall" statements so that each one can be verified by a test [1].

ID	Requirement
FR1	The system shall allow users to log in using their email address and password, and shall direct each user to the screens of their role (Client, Receptionist, or Manager).
FR2	The system shall allow new clients to register an account, and shall allow users to reset a forgotten password by email.
FR3	The system shall allow clients to view the available time slots for a selected service and date, and book an appointment.
FR4	The system shall prevent a time slot that is already booked from being booked again, and shall display the message "This time slot is already booked."
FR5	The system shall allow clients to modify or cancel their own appointments while the appointment status is Scheduled.
FR6	The system shall allow receptionists to view the daily schedule and search for clients by name.
FR7	The system shall allow receptionists to change the status of an existing Scheduled appointment to Completed, Cancelled, or No-Show, and shall reject status changes for cancelled or non-existing appointments.
FR8	The system shall send an automated email reminder to the client 24 hours before each scheduled appointment.
FR9	The system shall generate daily reports for managers showing total bookings, cancellations, and attendance rate.

2.2 Non-Functional Requirements

ID	Category	Requirement
NFR1	Performance	The system shall respond to 95% of user requests within 2 seconds under normal load.
NFR2	Security	The system shall require authentication for all pages except login, registration and password reset, and shall store passwords only in hashed form.
NFR3	Availability	The system shall be available 99% of the time during working hours.
NFR4	Scalability	The system shall support at least 50 concurrent users without violating NFR1.
NFR5	Compatibility	The system shall work on current versions of common desktop and mobile browsers.
NFR6	Usability	A first-time client shall be able to complete a booking in under 2 minutes without help or training.

2.3 Target Users and Their Needs

User Role	Description	Needs (Requirements)
Client	Individuals seeking services at small clinics, salons, or consultation centers.	• A simple and easy-to-use interface to book, modify, or cancel appointments. • The ability to manage their bookings without calling the center.
Receptionist	Staff members responsible for organizing the daily schedule.	• A clear digital view of all scheduled appointments to prevent double bookings and lost appointments. • Tools to update appointment statuses (Scheduled / Completed / Cancelled / No-Show) in real time.
Manager	Business owners or supervisors who monitor performance.	• Access to daily reports including total bookings, cancellations, and attendance rate. • The ability to monitor system activity to improve productivity and customer satisfaction.

2.4 User Personas

Persona 1: The Client (Nora)

Role	Client
Background	A working professional with a tight schedule who finds it inconvenient to make phone calls during business hours to book appointments.
Frustrations	Forgetting appointment details and long waiting times at clinics.
Goals	To book, modify, or cancel appointments easily at any time through a user-friendly digital platform, and to receive automated email reminders.

Persona 2: The Receptionist (Ahmed)

Role	Receptionist
Background	Works at a busy medical center and manages schedules manually using notebooks and messaging apps.
Frustrations	Double bookings, lost appointments, and the difficulty of tracking attendance rates.
Goals	A centralized digital system to manage the daily schedule efficiently, prevent double bookings, and update appointment statuses in real time.

2.5 Use Case Diagram

The use case diagram shows the three human actors and the external Email Service. All human actors must log in before using their functions; the Email Service receives the reminders that the system generates for scheduled appointments.

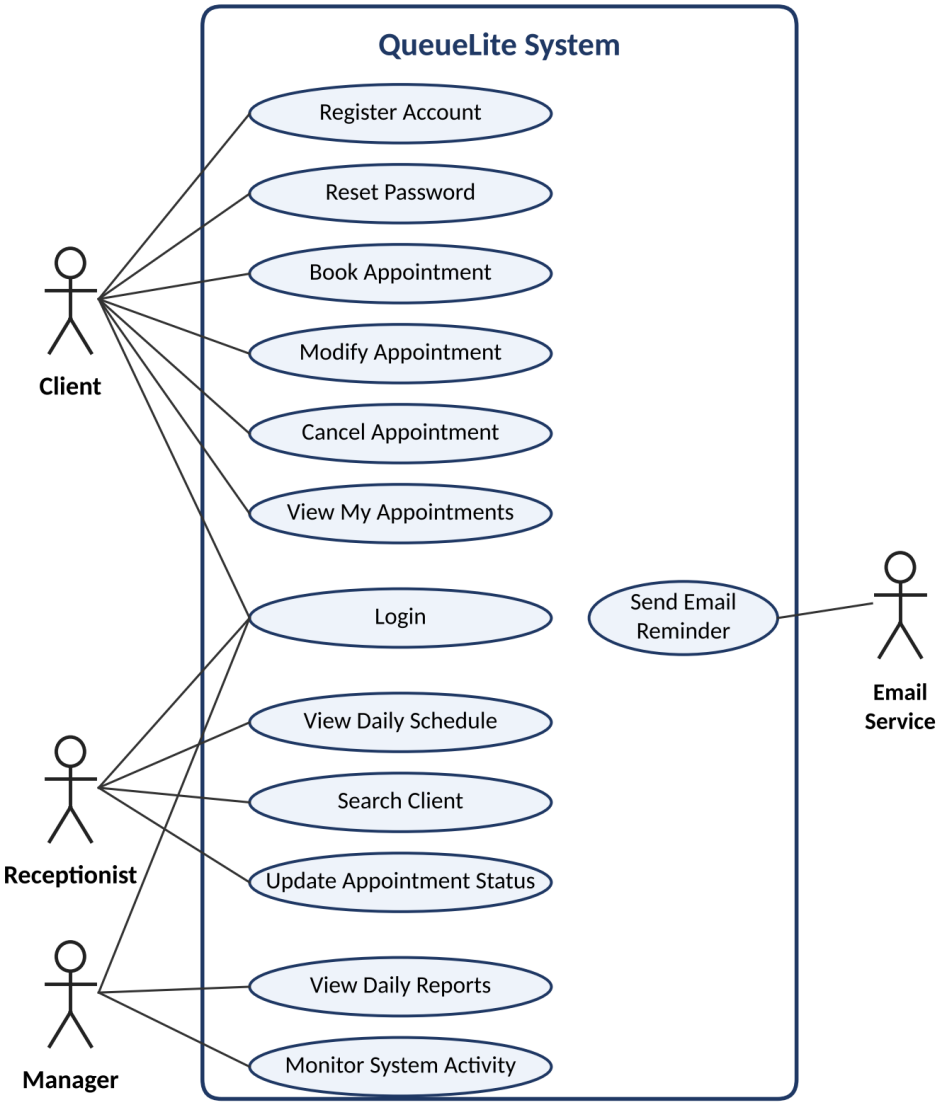


Figure 1: QueueLite use case diagram

2.6 Potential Challenges and Risks: Risk Assessment and Mitigation Plan

ID	Risk	Impact	Mitigation
R1	Double booking or scheduling conflicts	High – affects the reliability of the whole system	• Validate slot availability before confirming a booking (FR4) • Add a unique database constraint on the date and time slot to block overlapping appointments
R2	Email reminder failure	Medium – clients may miss appointments	• Log every sent and failed email and retry failed ones • Let clients view their upcoming appointments in the system at any time
R3	Incorrect system usage	Medium	• Validate inputs (e.g., required fields) • Use clear labels and error messages • Show confirmation prompts before cancelling or changing an appointment
R4	Unauthorized access	High	• Implement role-based access control • Store passwords hashed, never in plain text • Validate user sessions and expire inactive ones
R5	Delays in the project timeline	Medium	• Follow the weekly milestones strictly • Hold weekly progress meetings • Assign a backup member for each task

3. Design Stage

3.1 UML Class Diagram

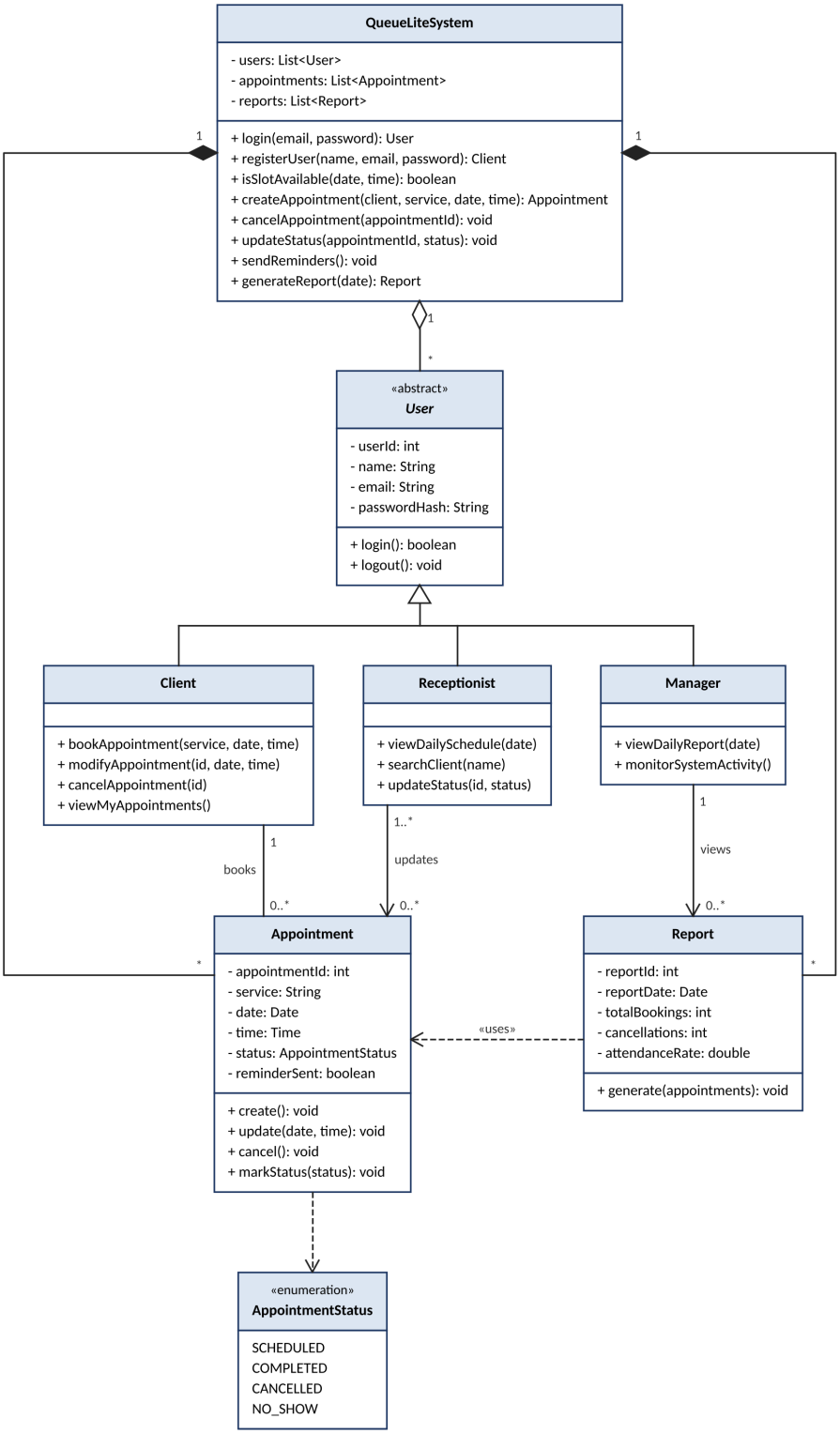


Figure 2: QueueLite UML class diagram

Key design decisions in the class diagram:

- **User** is an abstract class. **Client**, **Receptionist** and **Manager** inherit its attributes (userId, name, email, passwordHash) and its login/logout operations, so these attributes are not repeated in the subclasses.
- **QueueLiteSystem** holds the lists of users (aggregation), appointments and reports (composition). It performs the slot-availability check (isSlotAvailable) that prevents double booking.
- A **Client** books zero or more **Appointments**; each appointment belongs to exactly one client. **Receptionists** update appointment statuses and a **Manager** views **Reports**.
- Appointment status uses the **AppointmentStatus** enumeration (SCHEDULED, COMPLETED, CANCELLED, NO_SHOW) instead of free text, so invalid values cannot be stored. NO_SHOW is needed to calculate the attendance rate.
- **Report** summarizes appointments: attendance rate = Completed ÷ (Completed + No-Show) for the selected day.

3.2 System Architecture

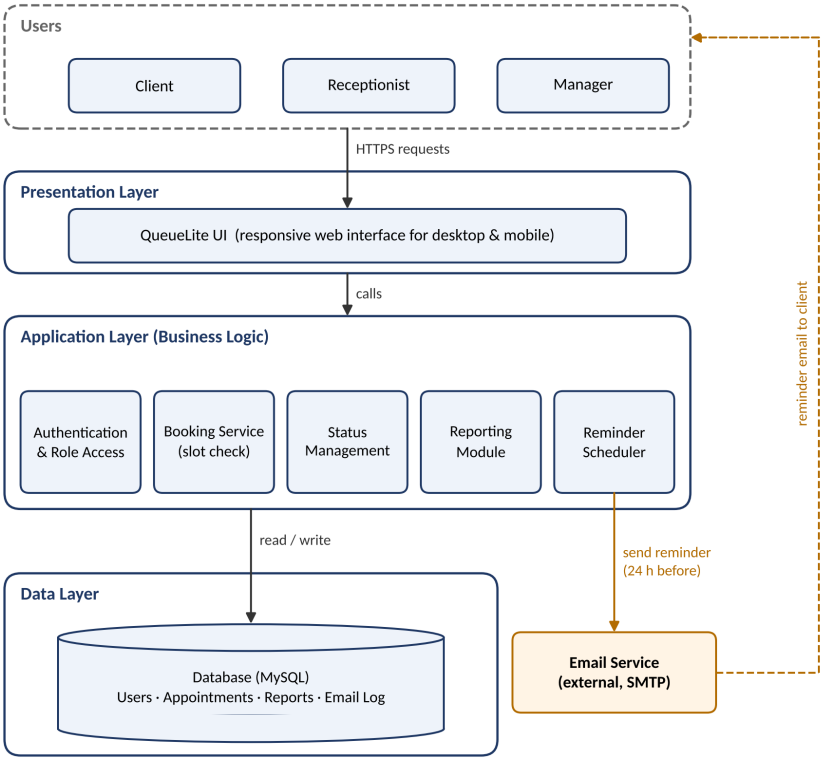
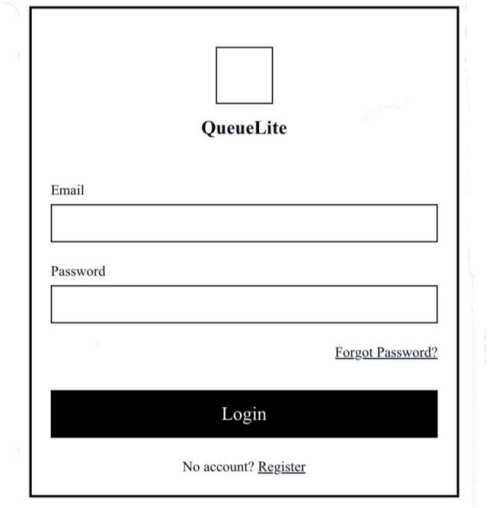


Figure 3: QueueLite three-layer system architecture

Clients, receptionists and managers use QueueLite through a responsive web interface (presentation layer). The application layer contains the business logic: authentication and role-based access, the booking service that checks slot availability, status management, reporting, and a reminder scheduler. The application layer reads and writes data in the database (data layer). The reminder scheduler sends reminders through an external email service, which delivers them to clients.

3.3 UI Wireframes

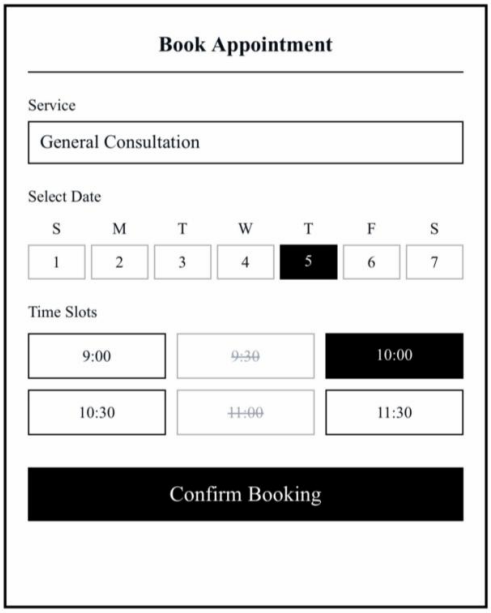
The wireframes were created with Wireframe.cc [4] and follow the requirements in Section 2.1.



Description

- Email and password input fields (FR1)
- "Forgot Password?" link for account recovery (FR2)
- Large Login button for easy use on mobile devices
- "Register" link for new clients (FR2)

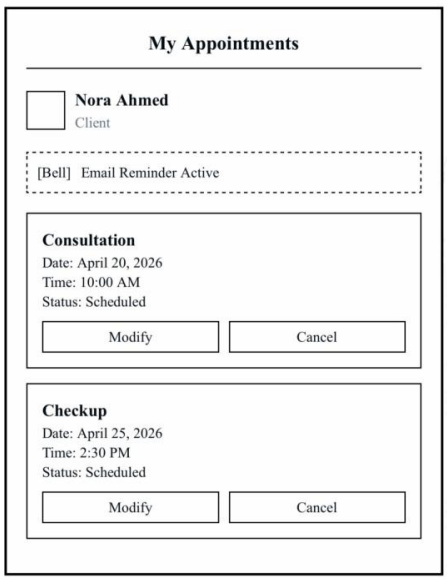
Figure 4: Login screen



Description

- Service selection field
- Calendar for selecting the date
- Time-slot grid; booked slots are shown with a strikethrough and cannot be selected (FR3, FR4)
- "Confirm Booking" button

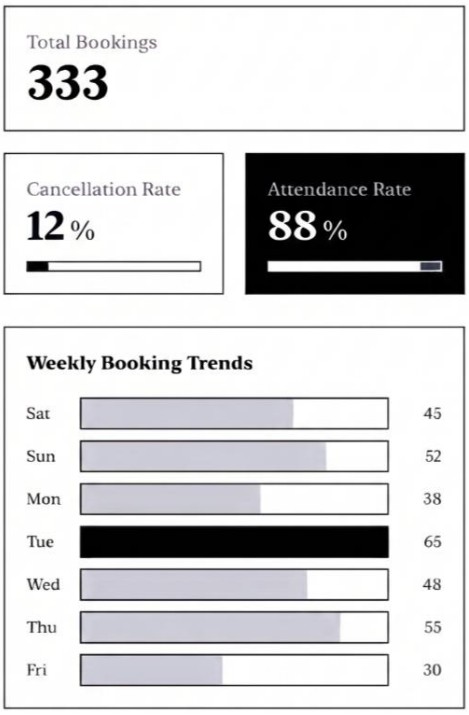
Figure 5: Book Appointment screen



Description

- Profile section showing the client's name
- "Email Reminder Active" banner (FR8)
- List of upcoming appointments with service, date, time and status
- "Modify" button to reschedule and "Cancel" button to cancel an appointment (FR5)

Figure 6: My Appointments screen (client Nora)



Description

- **Total Bookings:** number of bookings in the selected period (the example shows one week: 333 = sum of the daily values)
- **Cancellation Rate:** percentage of bookings that were cancelled
- **Attendance Rate:** Completed ÷ (Completed + No-Show) appointments; cancelled bookings are excluded
- **Weekly Booking Trends:** number of bookings on each day of the week; the daily report (FR9) is the same view for a single day

Figure 7: Manager reports dashboard

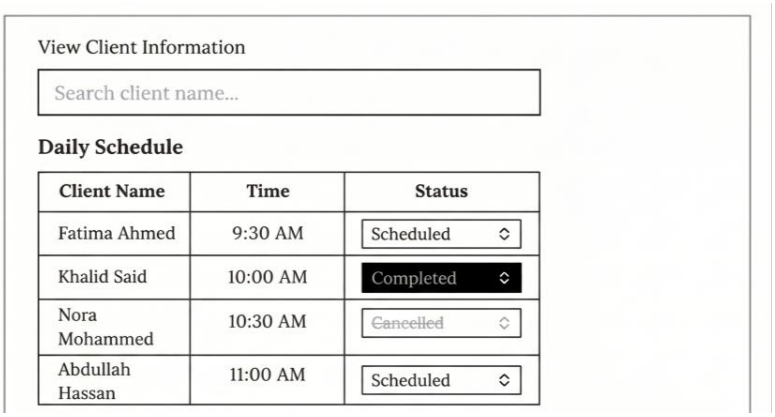


Figure 8: Receptionist daily schedule

Description

- **Client search:** find a client by name to view their information (FR6)
- **Daily schedule:** the day's appointments with time and status; the status is changed using the dropdown (FR7)

4. Testing Stage

4.1 Test Scenarios

ID	Scenario	Description
TS01	Client books an appointment	A registered client selects an available time slot and confirms the booking successfully.
TS02	Receptionist updates appointment status	The receptionist changes an appointment from Scheduled to Completed after the client attends.
TS03	Double booking is prevented	A second client tries to book a time slot that is already booked; the system must reject it.
TS04	Invalid status update is rejected	The receptionist tries to mark a cancelled or non-existing appointment as Completed; the system must reject it.

4.2 Test Cases

Test Case ID	**TC01**
Related Scenario	TS01 – Client books an appointment
Requirement	FR3
Description	Verify that a client can book an appointment using an available time slot.
Precondition	A client account exists; the selected slot is free.
Steps	1. Open the QueueLite system. 2. Log in as a client. 3. Select a service and a date. 4. Select an available time slot. 5. Click "Confirm Booking".
Expected Result	The booking is saved with status Scheduled and a confirmation message is displayed.
Actual Result	The appointment was booked successfully and a confirmation message appeared.
Status	**Pass**

Test Case ID	**TC02**
Related Scenario	TS02 – Receptionist updates appointment status
Requirement	FR7
Description	Verify that the receptionist can change an appointment status from Scheduled to Completed.
Precondition	A Scheduled appointment exists for today.
Steps	1. Open the QueueLite system. 2. Log in as a receptionist. 3. Open the daily schedule. 4. Select a Scheduled appointment. 5. Change the status to "Completed". 6. Save the changes.
Expected Result	The status is updated and the appointment shows "Completed".
Actual Result	The appointment status was updated successfully to "Completed".
Status	**Pass**

Test Case ID	**TC03**
Related Scenario	TS03 – Double booking is prevented
Requirement	FR4
Description	Verify that the same time slot cannot be booked by two clients.
Precondition	Two client accounts (Client A and Client B) exist.
Steps	1. Log in as Client A and book the 10:00 AM slot. 2. Log out and log in as Client B. 3. Select the same date and the 10:00 AM slot. 4. Click "Confirm Booking".
Expected Result	The second booking is rejected with the message "This time slot is already booked."
Actual Result	The system accepted both bookings for the same time slot.
Status	**Fail** Defect logged as **B01** (see Section 4.3)

Test Case ID	**TC04**
Related Scenario	TS04 – Invalid status update is rejected
Requirement	FR7
Description	Verify that a cancelled or non-existing appointment cannot be marked as Completed.
Precondition	A Cancelled appointment exists.
Steps	1. Log in as a receptionist. 2. Search for the cancelled appointment. 3. Change the status to "Completed". 4. Click Save.
Expected Result	The update is rejected with the message "Invalid appointment. Status cannot be updated."
Actual Result	The system updated the status without validation.
Status	**Fail** Defect logged as **B02** (see Section 4.3)

4.3 Bug Reports

Bug ID	**B01**
Title	System allows double booking for the same time slot
Severity	High
Priority	High
Found by / Related to	TC03 / FR4
Description	The system allows more than one client to book the same appointment time slot, causing scheduling conflicts.
Steps to Reproduce	1. Log in as Client A. 2. Book an available appointment slot (e.g., 10:00 AM). 3. Log in as Client B. 4. Select the same appointment slot. 5. Confirm the booking.
Expected Result	The system prevents the second booking and displays: "This time slot is already booked."
Actual Result	The system accepts both bookings for the same time slot.
Status	Open

Bug ID	**B02**
Title	Appointment status can be updated without a valid booking
Severity	Medium
Priority	High
Found by / Related to	TC04 / FR7
Description	The system allows the receptionist to change the status to "Completed" even when the appointment does not exist or has already been cancelled.
Steps to Reproduce	1. Log in as a receptionist. 2. Search for a cancelled or non-existing appointment. 3. Change the appointment status to "Completed". 4. Click Save/Update.
Expected Result	The system rejects the update and displays: "Invalid appointment. Status cannot be updated."
Actual Result	The system updates the status without validation.
Status	Open

References

[1] Sommerville, I. (2016). *Software engineering* (10th ed.). Pearson.

[2] Pressman, R. S., & Maxim, B. R. (2014). *Software engineering: A practitioner's approach* (8th ed.). McGraw-Hill Education.

[3] Oracle Corporation. (n.d.). *MySQL documentation*. https://dev.mysql.com/doc/

[4] Wireframe.cc. (n.d.). *Wireframe.cc: Minimal wireframing tool*. https://wireframe.cc/